Cameron J. Lerch

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Education

Yale University — Aug. 2019 – Aug. 2025
M.Sci Mechanical Engineering and Materials Science
M.Phil Mechanical Engineering and Materials Science
Ph.D. Coursework Completed; Advanced to Candidacy
NSF Graduate Research Fellow

Missouri University of Science and Technology — May 2019
B.S. Physics, Summa Cum Laude — GPA: 3.9/4.0
Minors: Mathematics, Philosophy, Geology

Experience

Oxylus Energy – Software and Mechanical Engineering Consultant — Aug. 2025 – Present
- Architected and implemented a modular PyQt5-based instrument control system to manage multi-station hardware, enabling per-station configuration, real-time telemetry, and synchronized schedule execution.
- Designed a scalable station-configuration framework supporting dynamic USB device mapping, live sensor polling, and safe schedule validation for laboratory automation.
- Designed and fabricated a sealed large-scale electrolysis stack enclosure with fume-hood exhaust, applying OSHA industrial ventilation guidance to ensure effective CO dilution and removal.

Yale University Graduate Researcher
Intelligent Autonomy Lab — Dec. 2021 – Aug. 2025
- Developed novel control algorithms for autonomous search and exploration tasks using multi-robot systems.
- Utilized optimization-based trajectory planning for multi-robot systems.
- Engineered safety-critical control frameworks using Control Barrier Functions for safe autonomous search and exploration in cluttered environments.
- Carried out real-world and simulated experiments using teams of drones.

O'Hern Research Group — July 2019 – Dec. 2021
- Researched glass-forming ability of bulk metallic glasses using LAMMPS for phase transition analysis.
- Engineered parallelized simulation workflows on HPC clusters for large-scale computation.
- Studied clogging of deformable particles in hopper flows using the soft particle model via LAMMPS (2^{nd} -3^{rd} year)

Yale University - Summer Undergraduate Research Fellow with Prof. Corey O'Hern — June 2018 – Aug. 2018
- Developed high performance molecular dynamics simulations of bulk metallic glasses using LAMMPS
- Calculated rearrangement statistics during cyclic deformation to characterize ductility

Missouri University of Science and Technology – Undergraduate Research Experience — Oct. 2016 – May 2019
- Developed parallel Monte Carlo simulations of diluted hexaferrites with Dr. Thomas Vojta.
- Performed quantum phase transition modeling using large-scale cluster computing with Dr. Thomas Vojta.
- Investigated orbital angular momentum using high powered lasers and optics with Dr. Daniel Fischer.
- Programmed simulation and analysis tools in MATLAB and Fortran.

Technical Skills

Python	MATLAB	C++	Fortran 90	ROS1/2	LAMMPS	Linux
Autodesk Inventor	SolidWorks		Siemens NX	Origin Graphing and Analysis		GUI Design

Publications D. Lee, **C. Lerch**, F. Ramos, I. Abraham, *Stein Variational Ergodic Search*, 2024, arXiv: 2406.11767

L. Sowadski, S. Anderson, **C. Lerch**, J. Medvedeva, T. Vojta, *Magnetic Properties of Diluted Hexaferrites*, Physical Review B (2024), arXiv: 2405.17328

A. Seewald, **C. Lerch,** M. Chancán, A. Dollar, I. Abraham, *Energy-Aware Ergodic Search: Continuous Exploration for Multi-Agent Systems with Battery Constraints*, 2024 IEEE ICRA, arXiv:2310.09470

C. Lerch, D. Dong, I. Abraham, *Safety-Critical Ergodic Exploration in Unknown Environments via Control Barrier Functions*, 2023 IEEE ICRA, arXiv:2211.04310

G. Khairnar, **C. Lerch**, T. Vojta, *Phase Boundary Near a Magnetic Percolation Transition*, Eur. Phys. J. B (2021), arXiv:2011.03390

J. Crewse, **C. Lerch** and T. Vojta, *Quantum critical behavior of a three-dimensional superfluid-Mott glass transition*, Phys. Rev. B 98 (2019), 054514

C. Lerch and T. Vojta, *Superfluid density and compressibility at the superfluid-Mott glass transition*, Eur. Phys. J. Spec. Top. (2019), arXiv:1712.08245

Presentations **C. Lerch**, D. Dong, I. Abraham, *Safety-Critical Ergodic Exploration in Unknown Environments via Control Barrier Functions*, 2023 IEEE ICRA, London, England (1 June 2023)

C. Lerch and C. O'Hern, *Encoding Multidirectional Memory in Sheared Amorphous Solids*, Leadership Alliance National Symposium, Hartford, CT (27 July 2018)

C. Lerch and T. Vojta, *Monte Carlo simulations of the magnetic behavior of diluted hexaferrites,* APS March Meeting, Los Angeles, CA (9 Mar 2018)

T. Vojta, J. Crewse, and **C. Lerch**, *Quantum critical behavior of a three-dimensional superfluid-Mott glass transition*, APS March Meeting, Los Angeles, CA (8 Mar 2018)

Honors & Activities

2019 National Science Foundation Graduate Research Fellowship Program (NSF-GRFP) recipient
2019 Missouri Undergraduate Research Day at the Capitol – selected participant
2018 Yale University CRISP Summer Undergraduate Research Fellow
47th Annual Missouri S&T Fuller Poster Competition - winner
Sigma Pi Sigma Physics Honors Society
Society of Physics Students
S&T Astronomical Research Society – former Vice President
2014 FIRST Dean's List Finalist